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Anomaly Detection in Industrial Quality Control with Computer Vision and Deep Learning

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***Abstract*—This study examines the impact of the dynamic manufacturing environment on companies' efforts to improve customer satisfaction through product quality enhancement, production cost reduction, and sustainability realisation. The research focuses on the current trends and challenges companies face in adapting in response to the dynamic nature of the industrial industry. This study seeks to offer practical recommendations by analyzing these factors for companies to meet the evolving demands of their customers while ensuring long-term success and competitiveness. This study examines the impact of anomaly detection on product quality, specifically focusing on the commonly employed method of visual quality inspection. This study examines the methods of visual quality inspection for products, comparing manual and automatic approaches. This study examines the limitations of manual inspection as a monotonous task, which can result in the oversight of errors and subjective assessments. In order to keep up with the needs of smart production and the advent of Industry 4.0, manufacturing as a whole is actively seeking computerized quality inspection solutions. Innovative inspection systems can be created through the effective use of big data. This research abstract discusses the widespread use of machine learning in automated quality inspection systems to enhance performance and support data analytics methods. This study uses a convolutional neural network architecture to extract complex features and patterns from images. This enables the model to differentiate between normal and abnormal conditions effectively. This research presents a proposed model trained on a diverse dataset, including a comprehensive collection of potential defects and irregularities. This research abstract presents a proposed framework for validating effectiveness through extensive experiments on datasets from diverse industrial domains. This study presents the findings of a research project that evaluated the performance of a deep learning model in detecting anomalies. The results indicate that the deep learning model outperformed traditional methods in terms of anomaly detection. This suggests that the deep learning model can effectively adapt to changing production environments.**

Keywords*—anomaly detection, convolutional neural network, dynamic manufacturing environment, product quality enhancement, sustainability.

I. Introduction

Rapid technology breakthroughs, shifting customer expectations, and the requirement for sustainable practices all contribute to the ever-changing world of modern production, which organisations must adapt to. Businesses strive to improve consumer satisfaction by several techniques, such as reducing production costs, improving product quality, and achieving sustainability goals [1-2]. This study explores the complex relationship between these aspects and how they affect these efforts. The goal is to provide practical solutions that promote long-term success and competitiveness by exploring current trends and difficulties organisations confront in this dynamic industrial terrain. Ensuring product quality is essential in the ever-changing production industry, which presents numerous problems. Companies are forced to embrace new strategies to meet and surpass the ever-increasing expectations of their customers. The importance of anomaly identification in product quality is the main topic of this study, which zeroes in on visual quality inspection as a specific technique. This study intends to illuminate the relative merits and shortcomings of various visual quality inspection strategies by investigating their approaches, including automated and manual procedures.

This study challenges the conventional wisdom by examining the drawbacks of manual inspection, including the potential for oversight errors and subjective assessments caused by its monotony. To meet the demands of smart manufacturing and the Industry 4.0 paradigm, the manufacturing industry is moving away from manual inspection and toward automated quality inspection systems [3-4]. An essential step in creating smart quality inspection systems that can adjust to the intricacies of contemporary manufacturing processes is the effective use of big data. In this context, automated quality inspection becomes a powerful tool for machine learning [5-9], particularly convolutional neural networks. These innovations make extracting intricate patterns and features from visual input possible, giving the system a leg up when distinguishing between typical and out-of-the-ordinary situations. In this study, we offer a model that has been trained on a large dataset that includes a wide variety of possible flaws and anomalies. These represent problems that people face in manufacturing settings.

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The conventional approach to inspection entails utilising human labour to identify and assess defects and deviations. The apparent shortcomings of manual inspections have led to an increasing dependence on algorithmic techniques from anomaly detection in the past few years.]. These drawbacks include the significant amount of time and effort required for manual inspections and the potential for errors. As a result, modern strategies are shifting towards algorithms to streamline the inspection process and improve accuracy [10]. Deep neural networks have garnered significant attention in machine and deep learning because of their capacity to understand intricate connections from extensive datasets [11-12].

In their study, Tamilselvan and Wang [13] employed deep belief networks as a means to detect and diagnose potential engine problems and power transformer breakdowns in aero planes. As part of their analysis and identification of cable stress patterns displayed by a cable-stayed bridge under typical usage, Wei et al. [14] used clustering methods. Evaluating gearbox faults, Li et al. [15] looked into the use of a multimodal deep support vector classification method. In a study conducted by Ye et al. [16], the researchers employed a vision-based approach to measure the dynamic displacement of a long-span bridge. The measurement was carried out using a pattern-matching algorithm, which allowed for accurate tracking and analysis of the bridge's movements. This research aimed to provide valuable insights into the behaviour and performance of long-span bridges, contributing to structural engineering and bridge monitoring. In their study, Koch et al. [17] conducted a comprehensive review of the application of computer vision techniques for defect detection and condition assessment of civil infrastructure. In a recent study, Cha et al. [18] employed CNNs to identify and detect structural crack damage based on image analysis. Using video data, Yang et al. [19] were able to identify vibration patterns with great spatial resolution. Yeum et al. [20] presented a new method for picture localization based on a platform of aerial vision sensors. Their research aimed to develop a technique to effectively identify and filter out the small proportion of relevant or interesting images.

According to Nayak et al. [21], the two most popular types of deep learning algorithms used for detecting anomalies in videos are supervised and semi-supervised learning. In the initial category, utilising convolutional neural networks (CNNs) that incorporate spatiotemporal layers, such as convolutional 3D or two-stream CNN, has proven effective in generating video descriptors. These descriptors offer valuable discriminative information, particularly when labelled data is accessible.

II. Materials And Method

Here the author details the datasets used to assess the algorithms utilized in this research. In this study, they introduce the proposed deep learning approach and various other methods that have been thoroughly investigated.

A. Dataset and pre-processing

The dataset under consideration pertains to Steel plate faults, as compiled by Semeion, a renowned institution specialising in the Research of Sciences of Communication. The dataset under consideration involves the analysis of a stainless-steel leaf, specifically focusing on its superficial faults. The dataset includes a total of 7 faults, which are detailed in Fig.1. The characterisation of each fault involves the analysis of 27 attributes and features that capture the geometric shape and contour of the fault.

It is necessary to preprocess the steel plate defects dataset to train the anomaly detection model using high-quality, consistent, and relevant data. This is an important research step. We looked for missing values in the features and attributes by thoroughly examining the dataset. It is critical to apply appropriate procedures, like imputation or elimination, to fill in the gaps in the data without affecting its integrity. Research frequently employs these methods to deal with missing data and guarantee an unbiased study. While removal involves not including instances with missing data in the study, imputation involves calculating the missing values using the existing data. Depending on the dataset's unique properties and the study goals, one of these methods—each with its own pros and cons—may be more appropriate. Researchers can successfully deal with data gaps and keep the data intact throughout analysis by meticulously employing these procedures.

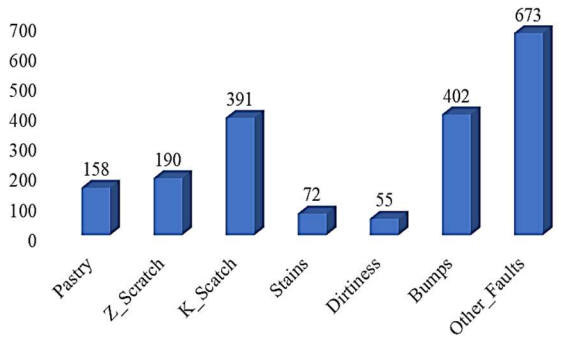

Fig. 1. plate fault classification from the dataset

$$Missing_{Values} = \sum_{i=1}^{N} \delta(x_i) \quad (1)$$

To avoid problems caused by differently sized characteristics, standardize their scales. At this point, we know that every feature is helping the model train to its full potential. Normalize the data to a standard scale (e.g., between 0 and 1) to make convergence during model training easier and account for measurement unit changes.

$$z_i = \frac{x_i - \mu}{\sigma} \quad (2)$$

where μ is the mean and σ is the standard deviation.

$$x_i' = \frac{x_i - min(X)}{max(X) - min(X)} \quad (3)$$

B. Model Architecture

The integration of CNNs and LSTMs has emerged as a robust architecture for anomaly detection in industrial quality control, particularly in sequential data scenarios. One notable advantage of CNN lies in its inherent ability to acquire the distinctive characteristics of the provided input data autonomously. This is achieved through the network's structured architecture; wherein various layers can learn distinct levels of features. Consequently, this hierarchical feature learning process significantly enhances object recognition accuracy. The CNN typically comprises three primary components: a data layer, a feature learning network, and a classification network.

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The data layer serves as the primary input for the network, encompassing either an unaltered image or a preprocessed image. The data provided by the user is transmitted to the learning features within the feature learning network. Subsequently, the acquired features undergo classification through the classification network. This study employed the CNN to generate input feature vectors for the LSTM model. Consequently, the classification network was eliminated, and the input vector for LSTM was chosen as the output vector from the final max-pooling layer. The Fig.2. shows the feature extraction using the CNN model.

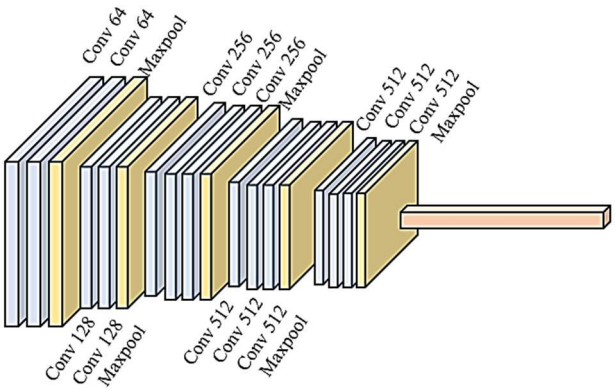

Fig. 2. Feature extraction using CNN model

Convolutional layers are a fundamental component of convolutional neural networks (CNNs) used in various computer vision tasks. These layers perform convolution, an essential operation in image processing and feature extraction.

$$C_i = \sigma(W_i * X + b_i), \tag{4}$$

where C_i is the output feature map, W_i is the convolutional filter, X is the input, b_i is the bias term, and $*$ denotes the convolution operation. The activation function being discussed is the Rectified Linear Unit (ReLU). One activation function that is frequently utilized in neural networks is ReLU. The input value is returned by a piecewise linear function if it is positive, and by zero if it is negative. Because of its ease of use and capacity to counteract the disappearing gradient, ReLU has become more popular in the past few years.

$$ReLU(x) = max(0, x) \tag{5}$$

Pooling layers are a common component in CNNs used for image recognition tasks. One popular type of pooling operation is the max pooling operation. Max pooling is a down-sampling technique that reduces the spatial dimensions of the input feature map.

$$P_i = MaxPool(C_i, pool_{size}) \tag{6}$$

where P_i is the pooled feature map, and $pool_{size}$ is the size of the pooling window. The batch normalisation operation is a technique used in neural networks to normalise the output of a previous activation layer. To do this, divide the output by the batch standard deviation after subtracting the phase mean. Because of this adjustment, the output now has a mean of 0 and a standard deviation of 1, which is good for the network's reliability and efficiency as a whole. This operation is typically applied during the training phase of the network and can be seen as a form of regularisation that aids in reducing the internal covariate shift.

The samples underwent feature extraction using CNN to obtain the equivalent feature vectors X. The feature vectors were sequentially fed into the LSTM model, a type of RNN. The LSTM processes the input vectors time-dependent, allowing it to capture temporal dependencies. The resulting outputs, denoted as O, represent the recognition results obtained from the LSTM model.

$$Input\ Features = [F_1, F_2, \ldots, F_t], \tag{7}$$

where t represents the sequence length, and F_i is the vector of features extracted by CNN for the i-th image.

$$h_t, c_t = LSTM(F_t, h_{t-1}, c_{t-1}) \tag{8}$$

where h_t is the hidden state, c_t is the cell state, F_t is the input feature vector at time t, and h_{t-1} and c_{t-1} are the previous hidden and cell states.

$$H = [h_1, h_2, \ldots, h_t] \tag{9}$$

representing the hidden states over the entire sequence.

$$F = Flatten(H) \tag{10}$$

where F is the flattened vector of LSTM hidden states.

$$O_i = \sigma(W_i \cdot F + b_i) \tag{11}$$

where O_i is the output of the dense layer, W_i is the weight matrix, F is the input vector, and b_i is the bias term. The LSTM architecture described in this study is designed to effectively handle sequential data derived from images in industrial quality control scenarios. It is specifically optimised to process features extracted by CNNs, enhancing its ability to handle such data. The model's effectiveness in detecting anomalies in time-series image data is enhanced by the combination of LSTM's capability to capture long-term dependencies and the feature extraction capabilities of CNNs. The model's comprehensive compilation and training process aims to enhance its proficiency in accurately discerning between normal and abnormal conditions within the industrial quality control domain.

To maximise its performance in industrial quality control anomaly detection, the hyperparameter tuning method for the proposed CNN-LSTM model systematically examined numerous configurations, as shown in Table I.

TABLE I. HYPERPARAMETER TUNING OF THE PROPOSED MODEL

Hyperparameter	Fine-tuned Values	Selected Values
CNN Layers		
Convolutional Filters	32, 64, 128	64
Convolutional Kernel Size	(3,3), (5,5)	(3,3)
Pooling Size	(2,2), (3,3)	(2,2)
LSTM Layer		
LSTM Units	64, 128, 256	128
Dropout Rate	0.2, 0.3, 0.5	0.3
Dense Layers		
Dense Units	64, 128, 256	128
Dropout Rate	0.3, 0.5, 0.7	0.5
Training Parameters		

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Learning Rate	0.001, 0.01, 0.1	0.001
Batch Size	32, 64, 128	64
Optimizer	Adam, RMSprop, SGD	Adam

After experimenting with 32, 64, and 128 convolutional filters in the CNN layers, 64 filters were the optimal number to achieve feature extraction balance. The decision to use a convolutional kernel size of (3,3) instead of (5,5) indicated a desire to capture more nuanced information. Also, the pooling size was adjusted between (2,2) and (3,3); to downsample while keeping important information, (2,2) was chosen. To strike a good compromise between model complexity and temporal dependence capture, we explored 64, 128, and 256 units in the LSTM layer and settled on 128 units. To regularise the LSTM and dense layers, a dropout rate of 0.3 was selected. A dropout rate 0.5 was chosen, and 128 units were preferred within the dense layers. To achieve steady convergence, the training parameters, including the learning rate, were fine-tuned among the following values: 0.001, 0.01 and 0.1. The optimal batch size was determined to be 64 to achieve a compromise between computational efficiency and model stability. The adaptive learning rate attributes of the Adam optimiser were considered superior to those of RMSprop and SGD. When all of these hyperparameters have been fine-tuned, the result is a configuration that has been carefully selected to detect anomalies in the particular industrial quality control setup.

III. Performance Evaluation

A. Performance Evaluation

Particularly in the setting of anomaly detection in industrial quality control, performance evaluation is an essential component of evaluating the efficacy of a machine learning model. How successfully the model detects outliers and differentiates between typical and unusual occurrences may be learned from the evaluation measures. Area under the receiver operating characteristic curve (AUC-ROC), recall, accuracy, precision, and F1 score are commonly used for binary classification applications like anomaly detection. Results from testing the proposed CNN-LSTM model on various defect types in the context of industrial quality control are shown in Table II. The model performs admirably across various error types when applied to industrial quality control. With a high accuracy (0.98), precision (0.98), recall (0.99), F1 score (0.98), and AUC-ROC (0.99) for Pastry defects, the model demonstrates its strong ability to detect these flaws with minimal false positives and negatives. In a similar vein, the model performs admirably when it comes to Z Scratch faults, with metrics like accuracy (0.99), precision (0.98), recall (0.99), F1 score (0.98), and AUC-ROC (0.99) demonstrating its power to differentiate Z Scratch errors from regular occurrences. Accuracy (0.98), precision (0.97), recall (0.99), F1 score (0.98), and AUC-ROC (0.99) all show that the model continues to perform well when applied to K Scratch problems, indicating that it can efficiently detect these faults while minimising false positives.

TABLE II. The Result of the Proposed Model

Fault Type	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Pastry	0.98	0.98	0.99	0.98	0.99
Z Scratch	0.99	0.98	0.99	0.98	0.99
K Scratch	0.98	0.97	0.99	0.98	0.99
Stains	0.98	0.98	0.99	0.98	0.99
Dirtiness	0.99	0.99	0.99	0.99	0.99
Bumps	0.97	0.97	0.98	0.97	0.98
Other Faults	0.98	0.98	0.99	0.98	0.99

A combination of the model's high accuracy (0.98), precision (0.98), recall (0.99), F1 score (0.98), and AUC-ROC (0.99) in stain detection further demonstrates its robust performance, demonstrating its reliability in accurately identifying stain cases. Impressive accuracy (0.99), precision (0.99), recall (0.99), F1 score (0.99), and AUC-ROC (0.99) are displayed by the model when it comes to correctly identifying Dirtiness issues. False positives and false negatives are well minimised. With a recall of 0.98, F1 score of 0.97, and AUC-ROC of 0.98, the system performs solidly in detecting Bumps; however, there is a slight decrease in precision, suggesting an additional improvement. Accuracy (0.98), precision (0.98), recall (0.99), F1 score (0.98), and AUC-ROC (0.99) all indicate that the model is highly effective in recognising Other Faults, which go beyond the defined categories. When detecting anomalies in industrial quality control, the model consistently outperforms other methods, regardless of the type of problem.

For the proposed CNN-LSTM model, the accuracy trends over 100 epochs are graphically shown in Fig. 3. This figure shows how well the model performed when learning new data and when tested on new data by differentiating between training and testing accuracy.

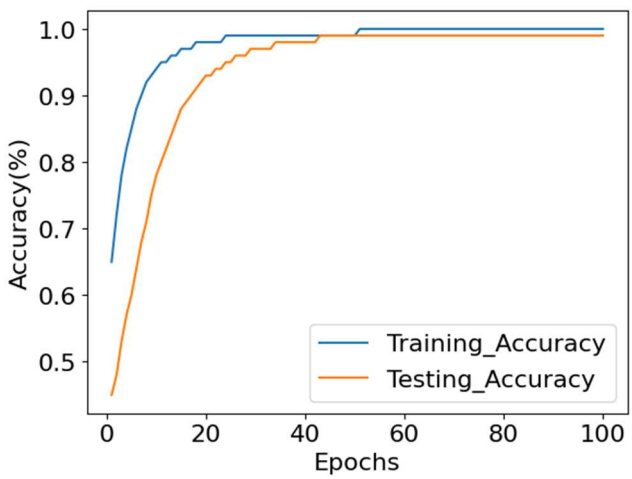

Fig. 3. Accuracy of the proposed model CNN-LSTM

The training accuracy curve illustrates the model's capacity to learn from the training data over successive epochs, which starts at 0.65 and steadily ascends to 1.00. An upward trend suggests that the training procedure was effective as the model iteratively adjusts its parameters to match the training dataset. When the model reaches a training accuracy of 1.00, it has fitted the training data perfectly, which

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could mean overfitting is possible. On the other hand, the learning curve for testing accuracy starts at a lower value of 0.45 and goes up to 0.99. If you look at this curve, you can see how well the model did on data it hadn't seen during training. This rising pattern indicates the model's ability to generalise to novel, previously unknown data. The fact that the testing accuracy is getting closer to 0.99 shows that the model is very accurate even when presented with examples it wasn't trained on. An essential factor is the difference between the training and testing accuracy curves. Overfitting, in which the model becomes too specific to the training data and fails to generalise well, may indicate a significant divergence, particularly towards the latter epochs. Conversely, suppose the two curves are closely aligned. In that case, it means the model is learning representative patterns from the training data and can still generalise to new data instances.

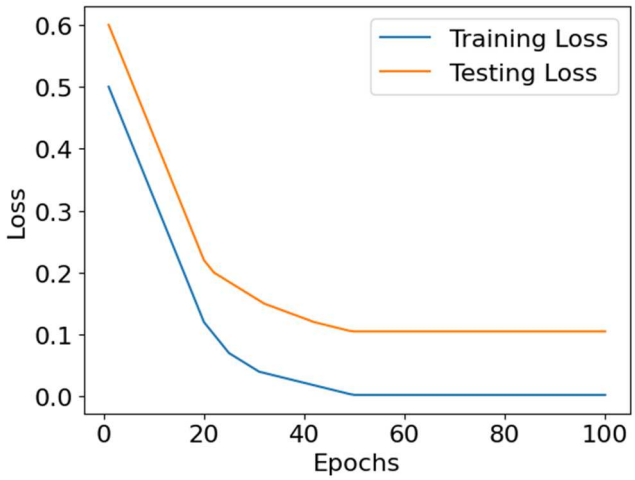

Fig. 4. Loss of the proposed model CNN-LSTM

Figure 4 shows the proposed CNN-LSTM model's training and testing loss dynamics across 100 epochs. The graph shows different patterns in the model's learning and generalizability to new data. The training loss curve starts at 0.5 and gradually drops to 0.0025, showing that the model better fits the training dataset and makes fewer mistakes as the epochs go by. The declining trend shows that the learning and convergence processes were successful, leading to a shallow training loss. With such a slight training loss, the model must have excellently adjusted to the complex training data. On the other hand, the testing loss curve starts at a higher value of 0.6 and then drops to 0.0050. This curve shows how well the model does on a different data set that wasn't used for training, showing that it can generalise beyond the instances shown. The fact that the testing loss is going down suggests that the model is still good at predicting outcomes in novel, unseen cases, which proves it can generalise effectively. As an exciting feature, the training and testing loss curves differ slightly. A close alignment indicates the model's ability to generalise and avoid overfitting the training data. On the other hand, overfitting—in which the model becomes excessively specific to the training dataset—may be indicated by a notable divergence, particularly in the latter epochs.

IV. Conclusion

Using a diversified dataset covering different types of faults, the given research concludes that a CNN-LSTM model can be used to solve the problem of anomaly identification in industrial quality control. The model consistently achieved high accuracy values throughout training and testing, showcasing its exceptional performance. Examining the model's performance in detecting certain anomalies in great detail reveals its effectiveness. Test results demonstrating consistently high accuracy values (0.45-0.99) highlight the model's ability to generalise to new data. The training accuracy indicated a satisfactory fit to the training dataset, reaching a perfect score of 1.00. However, it is worth cautiously considering that there is a tiny gap between the training and testing accuracy curves to prevent overfitting. The results show that the CNN-LSTM model is a good fit for automated anomaly identification in industrial quality control.

Improving the suggested CNN-LSTM model's capabilities will require a multi-pronged strategy for future work in anomaly detection for industrial quality control. Ensuring the model can adapt to different production conditions requires first working on extending the dataset and using advanced data augmentation techniques.

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